QQ-A-250/22A
June 27, 1974
SUPERSEDING
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QQ-A-00250/22(Navy - AS)
26 June 1969

FEDERAL SPECIFICATION

ALUMINUM ALLOY, 7178-T76, PLATE AND SHEET, ALCLAD (IMPROVED EXFOLIATION RESISTANT)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration for the use of all Federal Agencies.

(This specification forms a part of the latest issue of Federal Specification QQ-A-250).

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the specific requirements for improved exfoliation resistant alclad aluminum alloy 7178-T76 sheet and -T7651 plate. The general requirements are covered in QQ-A-250. The plate and sheet covered by this specification shall be an integral composite product consisting of a heat-treatable aluminum alloy (7178) core with thin layers of an aluminum alloy (7072) anodic to the core and of approximately equal thickness, bonded to both surfaces.

1.2 Classification. Sheet shall be furnished in the T76 temper only and plate shall be in the T7651 temper only, as specified (see 6.2). The definitions of these tempers, in accordance with ANSI H35.1, are as follows:

T76 - Solution heat-treated and artificially aged sufficient to produce required improved resistance to exfoliation and stress-corrosion cracking.

T7651 - Solution heat-treated, stress relieved by stretching to produce a permanent set of 2 percent nominal but not less than 1-1/2 nor more than 3 percent, and artificially aged sufficient to produce improved resistance to exfoliation and stress-corrosion cracking. Plate shall receive no further straightening after stretching.

FSC 9535

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2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

Federal Specification:

QQ-A-250	Aluminum and Aluminum Alloy Plate and Sheet; General Specification for

Federal Standards:

Fed. Test Method Std. No. 151	Metals; Test Method
Fed. Std. No. 184	Identification Marking of Aluminum, Magnesium and Titanium

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL-I-8950	Inspection, Ultrasonic, Wrought Metals, Process for

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

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American Society for Testing and Materials

ASTM D 1193	Standard Specification for Reagent Water
ASTM G 34	Standard Method of Test for Exfoliation Corrosion Susceptibility in 7XXX Series Copper Containing Aluminum Alloys (EXCO Test)
ASTM G 38	Recommended Practices for Making and Using the C-Ring Stress-Corrosion Cracking Test Specimen

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

American National Standards Institute Publication

ANSI H35.1	Alloy and Temper Designation System for Aluminum

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

3. REQUIREMENTS

3.1 Chemical composition. The chemical composition of the core ingots or slabs and of the cladding plates used for the manufacture of the clad plates and sheets shall conform to the requirements specified in table I, for core and cladding, respectively.

3.2 Mechanical properties. The mechanical properties perpendicular to the direction of final rolling, except for material less than 9 inches in width, shall conform to the requirements of table II for the temper specified. For material less than 9 inches in width, the mechanical properties parallel to the direction of final rolling shall conform to the requirements of table II for the temper specified.

3.3 Cladding thickness.

3.3.1 Thickness of cladding plates. The aluminum alloy (7072) plates that are bonded to the two sides of the aluminum alloy (7178) ingot or slab to form a composite that is to be rolled to the finished thickness shall each have a thickness not less than that specified in table III.

3.3.2 Average thickness of cladding. When specified (see 6.2), the average cladding thickness of the finished sheet or plate shall be determined. Samples, examined in accordance with QQ-A-250, shall show an average of cladding on each side as specified in table III.

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TABLE I. Chemical composition 1/

Element	Composition			
	Core (7178) Percent		Cladding (7072) Percent	
	Minimum	Maximum	Minimum	Maximum
Zinc	6.3	7.3	0.8	1.3
Magnesium	2.4	3.1	---	0.10
Copper	1.6	2.4	---	0.10
Chromium	0.18	0.35	---	----
Manganese	----	0.30	---	0.10
Iron	----	0.50	---	2/
Silicon	----	0.40	---	2/
Titanium	----	0.20	---	----
Others, each	----	0.05	---	0.05
Others, total	----	0.15	---	0.15
Aluminum	Remainder		Remainder	

1/ Analysis shall regularly be made only for the elements specifically mentioned in the table. If however, the presence of other elements is indicated in the course of routine analysis, further analysis shall be made to determine conformance to the limits specified for other elements.
2/ Iron plus silicon 0.7 percent maximum.

TABLE II. Mechanical properties

Form and Temper	Thickness in inches	Tensile strength, p.s.i. minimum	Yield strength at 0.2 percent offset, p.s.i. minimum	Elongation in 2 inches or 4 times D. percent minimum 2/ 3/
T76 1/	0.045-0.187	71,000	60,000	8
	0.188-0.249	73,000	61,000	8
T7651	0.250-0.499	72,000	60,000	8
	0.500-1.000	73,000	62,000	6

1/ Mechanical property requirements for sheet less than 0.045 inch in thickness are subject to inquiry.
2/ Not required for material 1/2-inch or less in width.
3/ D represents specimen diameter.

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TABLE III. Cladding thickness

Thickness of finished plate or sheet in inches	Nominal cladding thickness per side; percent of composite thickness	Average minimum cladding thickness per side on finished plate or sheet; percent of plate or sheet thickness
Under 0.063	4.0	3.2
0.063 thru 0.187	2.5	2.0
0.188 and over 1/	1.5	1.2

1/ For plate 0.500-inch and over, the average cladding thickness per side shall have a maximum value of 3 percent of the plate thickness.

3.4 Exfoliation and stress-corrosion cracking. Material in the T76 and the T7651 tempers when tested in accordance with 4.2.1 and 4.2.2 shall show neither exfoliation corrosion equivalent to or in excess of that depicted in Figure 2 of ASTM G 34 for Category B nor stress-corrosion cracking susceptibility beyond the acceptance criteria as detailed in 4.2.2.5.

3.4.1 Lot acceptance control criteria for T76 and T7651 tempers. Susceptibility to exfoliation and stress-corrosion cracking for each lot of T76 and T7651 temper material shall be determined by the following control criteria (see 4.1.2 and 4.2.2):

a. Determine mechanical properties and electrical conductivity.

b. If the conductivity is 38 percent IACS or higher, and the tensile properties meet the minimum limits herein, the material is acceptable.

c. If the conductivity is at least 35 percent IACS but less than 38 percent IACS, the material shall be tested as specified or reprocessed.

d. If the conductivity is below 35 percent IACS, the material is not acceptable and must be reprocessed.

3.5 Internal defects. Unless otherwise specified (see 6.2), plate shall be ultrasonically inspected (see QQ-A-250). Acceptance limits shall be as specified in table IV.

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TABLE IV. Ultrasonic discontinuity acceptance limits 1/

Thickness in inches	Maximum weight per piece in pounds	Discontinuity class 2/
0.500 thru 1.000	2,000	B

1/ Discontinuities in excess of those listed in table IV may be allowed subject to the approval of the procuring activity, if it is established that they will be removed by machining or that they are in non-critical areas.
2/ The discontinuity class limits are defined in MIL-I-8950.

3.6 Marking. In addition to the marking required in Fed. Std. No. 184, plate and sheet shall be identified by a lot number in at least one location on each piece.

4. QUALITY ASSURANCE PROVISIONS (See QQ-A-250).

4.1 Sampling.

4.1.1 For exfoliation and stress-corrosion cracking tests. Two samples shall be taken for each 4,000 pounds or less of the first three production lots for each size range of sheet and plate listed in table II. Thereafter, surveillance testing shall be performed on at least one sample per month for each size range of sheet and plate produced during that month which had previously been produced with acceptable exfoliation resistance. The surveillance test sample shall be taken from a lot which has met the requirements of 3.4.1(b). Samples as above shall be taken for stress-corrosion cracking test when thickness permit (see 4.2.2).

4.1.2 For mechanical property and conductivity tests. Sampling shall be in accordance with QQ-A-250 for mechanical property tests. Mechanical properties and conductivities shall be determined on the same samples (see 4.2.3). Conductivity shall be determined prior to testing for mechanical properties.

4.2 Tests.

4.2.1 Exfoliation corrosion test. Test shall be performed in accordance with ASTM G 34. Full thickness specimens of sheet and plate large enough to provide visual assessment of attack may be used; however, for all materials 0.100 inch or more in thickness, 10 percent of the thickness shall be removed by machining one surface. The machined surface shall be evaluated.

4.2.2 Stress-corrosion cracking test. Stress-corrosion cracking tests shall be performed on plate 0.75-inch and over in thickness as follows:

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4.2.2.1 Specimens. Specimens shall be selected in a manner such as to permit application of the specific tension stress in the short transverse direction. Specimens shall be C-Rings as described in Method 823 of Fed. Test Method Std. No. 151 or in ASTM G 38.

4.2.2.2 Exposure period. Exposure shall be for thirty (30) days.

4.2.2.3 Tension stress. The tensile stress applied in the short transverse direction (perpendicular to grain flow) shall be 25,000 p.s.i. and the specimen shall be held at constant strain.

4.2.2.4 Exposure procedure. The stressed specimen shall be exposed to a 3.5 percent salt solution prepared by dissolving reagent grade sodium chloride (NaCl) in distilled or deionized water, conforming to ASTM D 1193, Type II. The pH of the solution shall be 6.4 to 7.2. The solution temperature shall be 75 ±2°F, air temperature shall be 80 ±2°F, and with a relative humidity of 45 ±6 percent. The alternate immersion cycle shall consist of 10 minutes immersion in solution and 50 minutes in air, with sufficient circulation to dry the specimens slowly before the next immersion.

4.2.2.5 Stress-corrosion test results. After 30 day exposure, the specimen shall exhibit no visual evidence of stress-corrosion cracking. Any highly directional attack which is suspected of concealing a stress-corrosion crack shall be cross sectioned and examined metallographically. An example of stress-corrosion cracking is shown in Figure 1. Figure 2 illustrates pitting type attack which does not constitute failure.

4.2.3 Electrical conductivity tests. The conductivity shall be determined by taking three electrical conductivity readings at random on each sample. For thicknesses less than 0.100 inch, the conductivity shall be determined on the surface after removal of the cladding. For thicknesses of 0.100 inch and over, the conductivity shall be determined on the machined surface after machining one surface of the sample to a depth of approximately 10 percent of the product thickness. The surface machining may be accomplished by chemical milling. The average of three readings shall be used as the conductivity acceptance criteria of 3.4.1.

4.3 Records. The producer shall maintain records of the performance of all lots sampled and tested. Upon request of the procuring activity, such records shall be made available. These records shall be maintained for not less than one year after completion of the contract or purchase order.

5. PREPARATION FOR DELIVERY. (See QQ-A-250).

6. NOTES

6.1 Intended use. This alloy is intended for use where high strength is required and where exfoliation and stress corrosion resistance superior to 7178-T6 is required in high performance military aircraft.

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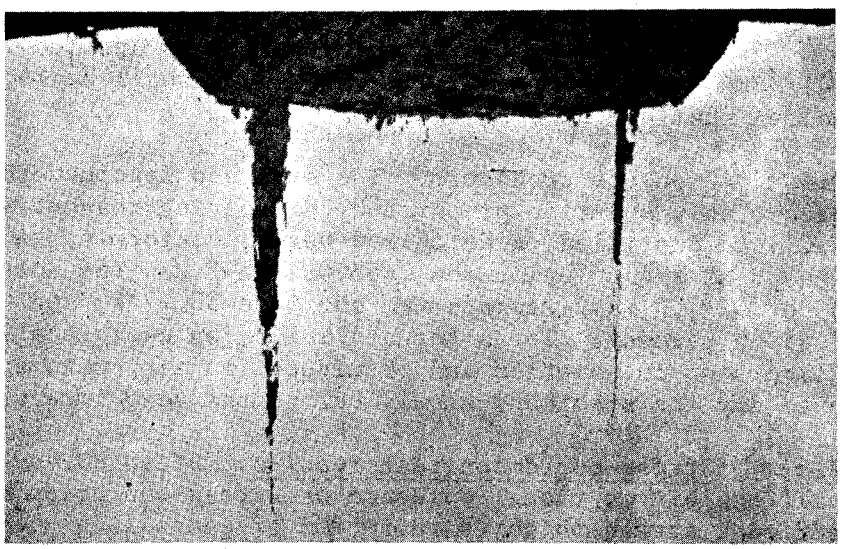


FIGURE 1

75X - UNETCHED STRESS CORROSION CRACKS
IN 7178-T7651 SPECIMEN. NOT ACCEPTABLE

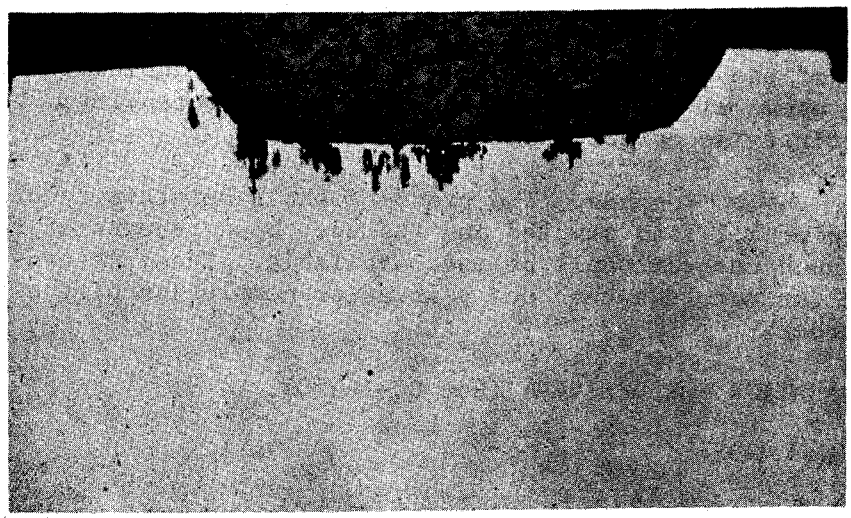


FIGURE 2

75X - UNETCHED PITTING TYPE ATTACK
IN 7178-T7651. ACCEPTABLE

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6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

a. Title, number, and date of this specification.

b. Form and temper required (1.2).

c. Dimensions required.

d. Requirements for sizes not specifically covered (see QQ-A-250).

e. Whether cladding thickness determination is required (see 3.3.2).

f. Ultrasonic inspection of material not required (see 3.5).

g. Selection of applicable levels of preservation, packaging, and packing required, if other than level C (see QQ-A-250).

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DOCUMENT IDENTIFIER AND TITLE
QQ-A-250/22A
ALUMINUM ALLOY, 7178-T76, PLATE AND SHEET, ALCLAD (IMPROVED EXFOLIATION RESISTANT)

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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